

Okay. I don't know. That's working. All right. Give me a longer stalk or something. No, no, it's fine. It's fine. Okay. So that session was about what doesn't work in AI yet, and obviously by implication, what might be done about it. So I'm going to talk about CASP, two things really, CASP results, which we use critical assessment to look at what's happening in AI in computational structural biology. This is fairly straightforward. Now that field is pretty much dominated by AI tools like AlphaFold. And so what is critical assessment telling us about where we've got to? But then I want to turn it around. In my world, critical assessment has always been this kind of community effort that you all know about, CASP, KG Dream, and so on, where the community does most of the critical assessment. I don't know. That's working. Just leave us even closer to the. All right. Give me a longer stalk or something. No, no, it's fine. It's fine. Okay. So that session was about what doesn't work in AI yet, and obviously by implication, what might be done about it. So I'm going to talk about CASP, two things really, CASP results, which we use critical assessment to look at what's happening in AI in computational structural biology. This is fairly straightforward. Now that field is pretty much dominated by AI tools like AlphaFold. And so what is critical assessment telling us about where we've got to? But then I want to turn it around. In my world, critical assessment has always been this kind of community effort that you all know about, CASP, KG Dream, and so on, where the community does most of the critical assessment. And the idea now is that for some things, you may be able to use AI, maybe not entirely, but at least as a new way of doing critical assessment. So that's the back arrow. I'm going to start with the CASP stuff and then flip it and go the other way, and I'll try and make that fit. Okay, so I think everybody knows probably about CASP. I'm not going to dwell on it. I just want to acknowledge my fellow organizers, and particularly Andre today, because the data I'm going to be showing you were worked up with him. And I'm going to go through, well, four in a little detail areas of computational structural biology, where the methods have got to, my speculation about why they're stuck. And then when we've done that, I'll go on to talk about other people's, of course, people aren't sitting on their hands here, they're trying to do something about it. What, as far as I know, people are going to try and do. Okay, so let's start with the one that works. And this is for single proteins. And some of you will have heard John Jumper talk about GDTTS last evening and trying to get the number up from 50 to 60. This is a main chain measure of accuracy. In these figures, green is usually experiment. Blue in a couple of occasions, purple, I think, is the calculated structure. This is over 1,000 residues. And you can see there's some domain swing here, but otherwise it's a really beautiful fit and not an easy target. So things really work, and that's reflected in this distribution of the best GDTTS from any participant in each of the last three CASPs. So 2020, 22, 24. And so in 2020, this was the introduction of AlphaFold 2. You can see the median was above 90 across all the targets. The average is a little bit lower because there were some failures. So notice this only goes down to 70. So we're just looking at the top end. And just to say, 90 is essentially what we think is beginning to close in on what you can expect in terms of agreement with

experiment. It varies with the experiment, uncertainty, environmental effects, all of those type of things. So you can see in 2022, CAS 15, there's a little bit of an improvement, no improvement in 2024. Now, this is not the AI methods failing as far as we can see. It's simply you're saturating. It's working, well not perfectly, you can see there's still some outliers, but in most cases, working very well indeed. So great. This one, for most senses, is done. But now we go on to other areas. First area I'm going to go to is protein assemblies, which we've actually been looking at for some CAS with our colleagues at Capri, and see how much deep learning has done so far there. So here's actually a 2024 target. This is a transmembrane protein. This is a hexamer packing against it. Very complex, lots of different interfaces, and beautifully modeled, colours as before. That wasn't the case before deep learning was introduced. So we've gone back again to 2020. People hadn't started using things like AlphaFold or RosettaFold. Go to 2022, those were introduced. A huge jump in accuracy. But this is actually an interface accuracy measure, and you can see we're going full 0 to 1 here. So we're not at the top end. Maybe a little bit of an improvement in 2024, but not much. And some of this may be with target differences. And if you look at the median, it's below 0.8. We don't, for interfaces, quite know what it should be, but it surely should be more than 0.8. I guess it's somewhere approaching 0.9, as with single proteins. So we've sort of saturated in two casps. Why have we saturated? Well, one popular reason is that there's less interface data to train on than there is for single proteins. A second reason, which I'll come back to, is that sometimes in interfaces you have less evolutionary information to guide how you match things up. And you can see, I'll come to an example where there was really failure there, and we'll see what people are doing about it. Again, these are my opinions. Some people might think differently. Let's go to RNA. So this is a new area in CASP. Here's a typical target. This is taken from the assessment paper by Rachel Kresch and her colleagues. And it's a very nice analysis. Let's just look at this one panel. What she's trying to look at here, each point is a RNA target structure. What she's looking at is to what extent the models are better than you could do from just taking the best template from the PDB. So if you're just copying templates, all the points will be on the diagonal. And you can see basically all the points are on the diagonal. And if so, mostly dominated by template performances, one exception. And when you go down to things with weak templates, that's down here, this is a different measure, TM-type measure of main chain accuracy. They're all bad. They're all less than 0.5. So two conclusions, not doing better than copying templates. When you haven't got a template, the methods are bad. And not apparent in what I'm showing you. The traditional models do as well as the deep learning models. So this is an area where so far deep learning has not had an impact. So why is that? Well, certainly, in this case, lack of training data. I don't know what the number is for RNA structures in the PDB, but it's probably, I don't know, less than a tenth, certainly, of the number of useful protein structures. And the other thing people suggest is the nature of the interactions here. You don't have 20 amino acids. You have these sort of relatively subtle changes in hydrogen bonding pattern. And particularly when you go to

unusual features, you tend not to pick up the right hydrogen bonding relationships. Again, this is speculation. Okay, my third example of where we've got to is for docking ligands, organic ligands, to proteins. And of course, this is very interesting in the context of designing drugs. And we really had, for the first time in 2024 in CASP, a reasonably good data set. So Michael Gilson was the assessor here. He got a couple of pharma companies to give us real-life sets. So these are structures which are determined in refining initial chemistries towards getting a final lead compound. And they're relatively easy compared with finding new chemistries, but still, as we'll see, not trivial. So I'm flipping scales on you. This is RMSD, so down is good. If we look, there is a lot of progress here. If we look at an old standard method, Auto-Doc Vinner, that's this blue bar here, you can see the RMSs are really bad. The average is about 5. If you go to the best CASP method, this is this Koziakov-Vager thing, the average is not great, it's 2.5, but it's certainly a huge improvement on this. And the median, lots of outliers, the median is lower at about 1.4 or so. Actually, the best performance here is a control, which is alpha-fold 3. And you can see here, you get to about one angstrom median, about just under 2 average. Now, what do you need? What is equivalent to experiment? What do you need not to do the experiment? Probably about 1 angstrom. And so in the best case here, in this relatively easy set, it's about 250 altogether, there is about half of them are not meeting the threshold of done. I think that's why you may have heard, you'll hear different talks in this conference about what people are trying to do about that, and we'll come back to it. Again, why? Lack of training data, good old sort of knee-jerk response. And in this case, a lot of different chemistries involved. So you've got 20 amino acids if you're folding a protein, or four bases if you're folding an RNA. Here, you may have a very large number of chemical groups. And somehow you have to train what environment those chemical groups actually would like. So I just want to mention about the outliers here. You can see even alpha-fold has some really bad results. And so the question is, are there some things here which just don't work? And just to make this one point, if you take this one and you look at what other people could do, you find people who actually got good RMSDs on that. So that's shown over here. It's a bit complicated. Again, the green here is the ligand, if you can see it. The blue is a prediction from some other group. And the purple is the deep mind prediction. You can see consistent with the RMS, it's really off, whereas somebody else got it right. So it's not that these things are sort of undoable, it's just that you sort of miss sometimes. Okay, there's another area of CASP I don't have data in for. We're trying to get away from one sequence, one structure, and so look at ensembles. Some of you may be aware there's a lot of controversy in the literature about how well that works with things like AlphaFold. The answer in our hands is it works spectacularly now and again, but generally doesn't work. So not the data for that, but it's another area where something needs to be done. All right, so that's 15 I'm in or what? Okay, so 15, okay, I'll speed up a bit. I enjoy talking about this stuff as the problem. Okay, so that's the story. What are people doing about it, as far as I know? And some people may know better than me. So the first solution, I've got

three things people are doing. The first is the knee-jerk one, let's get more training data. And you've heard this, I'm sure, a lot of times today. It's not clear to me when that's going to work and when it isn't. I guess for protein ligand, where the biggest effort is going, and that's the top here, it may work. So you're probably aware of some of these. You may have heard about them today, I'm not sure. OpenBind is getting 500,000 structures, at least that's the goal, to try and improve training data. Target 3035 is doing 2 trillion binding assays. They're not all structure, of course, but there'll be a lot of data. And OpenForm 3 is working with pharma companies to train on their internal data without it being exposed. And so huge amount of effort going into this. It will take some time. I don't know if it's going to work or not. Could you do the same with RNA? I asked Riju Dais, who knows about these things. And he said, to get enough training data, you probably want 10 to 100,000 structures costing somewhere between 100 million and 1 billion. So that's probably not going to be the answer there if there is an answer. You've got to find something else other than dumbly, I don't want to be rude to experimentalists, I was one once, generating more data. You might be able to do synthetic data for protein ligands. It's been suggested you might use molecular dynamics to get a training set, which I think is pretty cheeky of the molecular dynamics people, but we'll see. And there could be a possibility of bootstrapping. So a lot of things being tried. In some sense, a very simple idea, throwing money at the problem sometimes might work. What else could you do? So the thing I like, which there's not much evidence for, but I'll tell you what there is, is combining the deep learning methods with classical methods. So there's decades of development of methods not competitive on their own with deep learning, but certainly not garbage. And so if you could combine the two, you might get a better performance. And we saw a hint of that in this CASP. This is for difficult complexes, the ones that are in those outliers for protein complexes, where you don't have evolutionary information defining the interface. This is a nano-body antigen complex, and you can see it's beautifully predicted. It's this one here, and the best method that did this structure is Koziakov-Vadger. You can see all of the other methods, ColabFold, two versions of alpha-3 and so on, do not work. So what did they do? They had an FFT physics-based method which tries to search for poses. And so they ran that, they generated a set of poses, and then they used those as input to basically alpha-fold 3. And so you could regard it as a way of sampling, but it's a very intelligent way of sampling. It doesn't work every time yet. You can see there are still actually three failures. These are the set of targets we had for this class of problem. But I think there's a lot of hope here, and we saw a hint the other way around, do the physics afterwards for RNA. And so I know a lot of people are working on this, and I'd be surprised if there isn't something substantial in a year or two. Now, the third thing you could do, of course, is try a better algorithm. And there are a lot of ideas out there. So it could be the algorithm, it could be components, it could be the network setup. So lots, of course, you're familiar with this, convolution, graphs, transformers, and so on, represent atoms, residues, graphs, interaction types. All of these possibilities are being explored if you go to the literature.

And so all you've got to do is read the literature, get a good idea, and apply it, and win the next CASP. The problem is that if you go to the literature for deep learning, it's just proteins now. In the last five years, there are just under 3,000 papers. So we could do a little quiz. I guess Gustavo's got a quiz of some sort later on, as to whether or not you could tell me what's in each one of these papers, even if you were a specialist. And of course, you can't. So we're overwhelmed with ideas, some of which are just ideas, some of which are really substantiated by what's in the papers. But notoriously, you cannot rely on the peer review of these papers to tell you what to believe. You need a different way of doing critical assessment. And here's where I flip to the other arrow. How can you use AI, in this first example, and we're going to get to a second one, how can you use AI to critically assess the papers? And so I don't have many slides in for this, but just to give you the idea, the idea is that we define a set of criteria we'd like to be fulfilled. And then we ask people, me, to try and fill in those criteria. And then we ask different AI tools, nothing fancy here. We use sort of three central AI tools for this work, very amateur-ish, to do the same thing and see how well you can do. So here's an example using actually just ChatGPT with a lot of prompt engineering, of course, to try and look at it. It's an old paper. It's a diffusion docking, ligand docking paper. And the things we're looking at is what's the paper about? That's sort of easy. And there is a score for all this. Is this something which is interesting? Yes. Are there new ideas here? Yes, there are nice ideas. Is the algorithm well described? Yes, but some of the stuff may be inaccessible. But it is on GitHub, so that's good. Is the data available? Well, no, not really. And there's some doubts, I'm not shown it on here, about the training-testing distinction. So there's only a one for that. Is there a server? No, maybe that doesn't matter. Is it innovative? Yes, it's very innovative. So the idea is you can run this against the whole of the literature, not do it by hand. And you come to a paper, you want to say, well, what does the AI, or AIs preferably, think about this paper, get a quick impression of how seriously the method has been tested, whether it's a good idea, a novel idea relative to the background. Now, that I think would be useful, but it's not enough. What you'd want on top of that, I think, is some kind of dynamic ontology. What's out there, what's been looked at, how much, with what sort of methods. And so this is a sort of Mickey Mouse demonstration of principle on the next slide. I took the CASP abstracts. There's about 150 of them. They're very rough. We asked people to submit abstracts. They're not in any way peer-reviewed or standard or anything. And said, what methods are in these abstracts? And so this is a simple breakdown of that. It's sort of boring sometimes. Look, a lot of people use deep learning. What a surprise. But on the other hand, a lot of people use different sorts of sampling. And that's interesting because you know sampling is helping in a lot of these problems. So that's your top level of the ontology. Now you say, okay, what do you mean sampling? I would have a table in here if I'd put the slide in. What sorts of sampling are people doing? And then you could cross-correlate that. Of course, the abstracts don't contain the results. But you could cross-correlate that with the results and say, this sort of sampling tends to give a better result than this other sort of

sampling. So that's the concept. It's very much a concept, but I think it's extremely doable. Okay, so this is all within computational structural biology. But I think a lot of us, we've just heard a virtual cell talk, are interested in other sorts of assessment. And so one I've gotten very interested in is the problem of critical assessment in preclinical biology. So this is a bit of a jump. Let me introduce it with a slide. If there's a so-called reproducibility crisis in that field, and there's been careful studies to try and see how big a problem that is. So the idea is you take respectable papers, you look what experiments are in them, what the conclusions were, and then you try very hard, not just, you know, tried it didn't work stuff. Can I reproduce the reported result? So this is from this very nice large-scale study 2021, and you're looking at two things here. How well does the size of the effect which is reported reproduce? Let's say it's a five-fold change in expression of some gene. This is cancer stuff in some cancer cell. Is that reproduced? And secondly, is it significant by this traditional, very poor 0.05 probability criterion? So how much is that reproduced? So along the bottom here, you've got the reported cell size in log 2, so 2 to the 5 here. This is the replicated size. Unlike the RNA, the points should be on the diagonal here, and you can see they're not. They're more or less along, nothing happened. So startlingly, in these data, the average reported effect size in the reproduction is 20% of the original. And even if the statistical significance holds up, that takes you from biologically significant very often to biologically irrelevant. And then the colours here, the blue, you do replicate with significance, the orange, not surprisingly given where they are, you've lost your 0.05 significance. A few things actually flip. Here's something with quite a big change, but the opposite sign to what it was originally, and confidently not the right way around. So that's the problem this field has. And I would argue that's having a big impact on how well we can use all of these data. There's lovely, rich data out there, but nobody knows what to trust. And I'll show you one example of that to make the point. We're interested in one particular variant that affects risk of Alzheimer's disease. We'll go into a lot of specifics, but it's a missent single base change in this gene ApoE. So this is called ApoE4. If you look at the risk of getting Alzheimer's, most people are E3E3, and so this is for different alleles, the chance of not having Alzheimer's as a function of age. The black line is E3E3, the far red line here is E4E4. So you can see just there's a huge increase in risk here. Now a couple of points. If you understood why that increase was there, you could perhaps design prophylactics just for people with this variant. You might also be able to design prophylactics for people as you watch biomarkers over the years and say, because I know what's going on here, if I give you this drug, I can hold off Alzheimer's until you're dead, sort of thing. So that's what you would like to do. It's not an original thought. There are 7,000 papers on the E4 variant and risk of Alzheimer's. So it's worse than deep learning. So, oh, by the way, these are my colleagues on this project. So we did this first by hand. We sort of went through the 7,000 papers, and we said, we're thinking in mechanism terms here, what mechanisms have been proposed to explain this relationship. Now, I think what we know about other genetic variants in disease is that typically there's one mechanism. These systems are very robust,

of course, and so usually just changing one base may give you one way to some kind of disease phenotype. But here, if you go to the literature, you find, it could be three. If you go to the literature and look, this is us now, here are 30, actually I think there's only 28 on the plot, but it goes on beyond 30, different biologically discrete mechanisms which are in the literature supported by data. So there's not speculation here. There's a paper, it says experiment, experiment, experiment, conclusion. And what are you going to believe here? At least 90%, probably more, are wrong in some serious sense. How do you decide what's wrong and what's right? By the way, what I'm going to tell you by hand works. So you see there are big grances here. This is sort of classical amyloid stuff. There's a lot of role for lipids, perhaps. There's this other protein involved in Alzheimer's called tau. So we've been able pretty much by hand to eliminate the tau lot. We're very doubtful about the lipid lot. We're absolutely sure there is some mechanism involving amyloid. Believe you me, that's not an obvious conclusion in the field. And that's from doing the sort of thing I'm about to describe. What we do is to go the same kind of line as with the AI for looking at papers. We have a set of structured evidence criteria, and hopefully, yes, here's an outline of that. So they fall into four classes. There's often extrapolation involved in these experiments. So how valid is the extrapolation? Are you in a different organism, in vitro, in a cell, whatever? Statistics are where most of these papers fall down. I think that, not disrespecting my biology colleagues, they often don't have a good grasp of statistics, and the analysis is not valid, and that's where you get that sort of effect I showed you on the non-reproduction. On the other hand, some of these have been reproduced by different people. Some of them fit very well into other kinds of considerations of the biology. So you can plod through these rules by hand, you can score them, you can come up with a confidence. But you can also do it, again, with the AI tools. And I guess I'm about to run out of time. So this is a summary of trying to do that with this particular problem. There are different problems. Can you find the papers? Can you get the key information? Can you extract key results? Can you deduce the mechanism implications? Can you build a full mechanism? Can you critically assess the experimental results? Can you propose missing experiments? How easy is it to use? Generally low. In our hands, it takes forever, but this is obviously clearly cleanable, upable, and automatable on a large scale. I put probably better than people here. Actually, I'm convinced now it's a lot better. This is an old slide. So in my mind, we could take the whole of the biomedical literature, we could search for experiments, we could assess the reliability of the experiments, we could extract mechanism automatically, we could make a mechanism knowledge graph and find new mechanisms. That's the sort of vision here. It's obviously a ways to go with development, but I think that's really the future. And here comes the chair. I've done. One minute. I finished. I don't know what to say for a minute. Seriously, I finished. So we are actually in one hour break. Just to remind you, the title question for John. So we're going to work on the break. Actually, go for it. 40 minutes. So please come back. Listen to Justin and AI and the result of the. Thanks so much. That was great. Thank you. No, it's all time. It was great to see

you. I saw you in the AlphaPole video. And the idea now is that for some things, you may be able to use AI, maybe not entirely, but at least as a new way of doing critical assessment. So that's the back arrow. I'm going to start with the CASP stuff and then flip it and go the other way, and I'll try and make that fit. Okay, so I think everybody knows probably about CASP. I'm not going to dwell on it. I just want to acknowledge my fellow organizers, and particularly Andre today, because the data I'm going to be showing you were worked up with him. And I'm going to go through, well, four in a little detail areas of computational structural biology, where the methods have got to, my speculation about why they're stuck. And then when we've done that, I'll go on to talk about other people's, of course, people aren't sitting on their hands here, they're trying to do something about it. What, as far as I know, people are going to try and do. Okay, so let's start with the one that works. And this is for single proteins. And some of you will have heard John Jumper talk about GDTTS last evening and trying to get the number up from 50 to 60. This is a main chain measure of accuracy. In these figures, green is usually experiment. Blue in a couple of occasions, purple, I think, is the calculated structure. This is over 1,000 residues. And you can see there's some domain swing here, but otherwise it's a really beautiful fit and not an easy target. So things really work, and that's reflected in this distribution of the best GDTTS from any participant in each of the last three casps. So 2020, 22, 24. And so in 2020, this was the introduction of AlphaFold 2. You can see the median was above 90 across all the targets. The average is a little bit lower because there were some failures. So notice this only goes down to 70. So we're just looking at the top end. And just to say, 90 is essentially what we think is beginning to close in on what you can expect in terms of agreement with experiment. It varies with the experiment, uncertainty, environmental effects, all of those type of things. So you can see in 2022, CAS15, there's a little bit of an improvement. No improvement in 2024. Now, this is not the AI methods failing as far as we can see. It's simply you're saturating. It's working, well, not perfectly. You can see there's still some outliers, but in most cases, working very well indeed. So great. This one, for most senses, is done. But now we go on to other areas. First area I'm going to go to is protein assemblies, which we've actually been looking at for some CAS with our colleagues at Capri, and see how much deep learning has done so far there. So here's actually a 2024 target. This is a transmembrane protein. This is a hexamer packing against it. Very complex, lots of different interfaces, and beautifully modeled, colours as before. That wasn't the case before deep learning was introduced. So we've gone back again to 2020. People hadn't started using things like AlphaFold or RosettaFold. Go to 2022, those were introduced. A huge jump in accuracy. But this is actually an interface accuracy measure, and you can see we're going full 0 to 1 here. So we're not at the top end. Maybe a little bit of an improvement in 2024, but not much. And some of this may be with target differences. And if you look at the median, it's below 0.8. We don't, for interfaces, quite know what it should be, but it surely should be more than 0.8. I guess it's somewhere approaching 0.9, as with single proteins. So we've sort of saturated in two casps. Why have we saturated? Well, one popular reason is



that there's less interface data to train on than there is for single proteins. A second reason, which I'll come back to, is that sometimes in interfaces you have less evolutionary information to guide how you match things up. And you can see, I'll come to an example where there was really failure there, and we'll see what people are doing about it. Again, these are my opinions. Some people might think differently. Let's go to RNA. So this is a new area in CASP. Here's a typical target. This is taken from the assessment paper by Rachel Kresch and her colleagues. And it's a very nice analysis. Let's just look at this one panel. What she's trying to look at here, each point is a RNA target structure. What she's looking at is to what extent the models are better than you could do from just taking the best template from the PDB. So if you're just copying templates, all the points will be on the diagonal. And you can see basically all the points are on the diagonal. And if so, mostly dominated by template performances, one exception. And when you go down to things with weak templates, that's down here, this is a different measure, TM-type measure of main chain accuracy. They're all bad. They're all less than 0.5. So two conclusions, not doing better than copying templates. When you haven't got a template, the methods are bad. And not apparent in what I'm showing you. The traditional models do as well as the deep learning models. So this is an area where so far, deep learning has not had an impact. So why is that? Well, certainly, in this case, lack of training data. I don't know what the number is for RNA structures in the PDB, but it's probably, I don't know, less than a tenth, certainly, of the number of useful protein structures. And the other thing people suggest is the nature of the interactions here. You don't have 20 amino acids. You have these sort of relatively subtle changes in hydrogen bonding pattern. And particularly when you go to unusual features, you tend not to pick up the right hydrogen bonding relationships. Again, this is speculation. Okay. My third example of where we've got to is for docking ligands, organic ligands, to proteins. And of course, this is very interesting in the context of designing drugs. And we really had for the first time in 2024 in CASP a reasonably good data set. So Michael Gilson was the assessor here. He got a couple of pharma companies to give us real-life sets. So these are structures which are determined in refining initial chemistries towards getting a final lead compound. And they're relatively easy compared with finding new chemistries, but still, as we'll see, not trivial. So I'm flipping scales on you. This is RMSD, so down is good. If we look, there is a lot of progress here. If we look at an old standard method, Auto-Doc Vinner, that's this blue bar here, you can see the RMSs are really bad. The average is about 5. If you go to the best CASP method, this is this Kosiakov-Vager thing, the average is not great. It's 2.5, but it's certainly a huge improvement on this. And the median, lots of outliers, the median is lower at about 1.4 or so. Actually, the best performance here is a control, which is alpha-fold 3. And you can see here, you get to about one Angstrong median, about just under 2 average. Now, what do you need? What is equivalent to experiment? What do you need not to do the experiment? Probably about 1 angstrom. And so in the best case here, in this relatively easy set, it's about 250 altogether, there is about half of them are not meeting the threshold of done. I

think that's why you may have heard, you'll hear different talks in this conference about what people are trying to do about that, and we'll come back to it. Again, why? Lack of training data, good old sort of knee-jerk response. And in this case, a lot of different chemistries involved. So you've got 20 amino acids if you're folding a protein, or four bases if you're folding an RNA. Here, you may have a very large number of chemical groups. And somehow you have to train what environment those chemical groups actually would like. So I just want to mention about the outliers here. You can see even alpha-fold has some really bad results. And so the question is, are there some things here which just don't work? And just to make this one point, if you take this one and you look at what other people could do, you find people actually got good RMSDs on that. So that's shown over here. It's a bit complicated. Again, the green here is the ligand, if you can see it. The blue is a prediction from some other group. And the purple is the deep mind prediction. You can see consistent with the RMS. It's really off, whereas somebody else got it right. So it's not that these things are sort of undoable. It's just that you sort of miss sometimes. Okay, there's another area of CASP I don't have data in for. We're trying to get away from one sequence, one structure. And so look at ensembles. Some of you may be aware there's a lot of controversy in the literature about how well that works with things like AlphaFold. The answer in our hands is it works spectacularly now and again, but generally doesn't work. So not a lot of data for that, but it's another area where something needs to be done. All right, so that's 15 I'm in or what? Okay, so 15, okay, I'll speed up a bit. I enjoy talking about this stuff as the problem. Okay, so that's the story. What are people doing about it, as far as I know? And some people may know better than me. So the first solution, I've got three things people are doing. The first is the knee-jerk one, let's get more training data. And you've heard this, I'm sure, a lot of times today. It's not clear to me when that's going to work and when it isn't. I guess for protein ligand, where the biggest effort is going, and it's the top here, it may work. So you're probably aware of some of these. You may have heard about them today. I'm not sure. OpenBind is getting 500,000 structures, at least that's the goal, to try and improve training data. Target 3035 is doing 2 trillion binding assays. They're not all structure, of course, but they'll be a lot of data. And OpenForm 3 is working with pharma companies to train on their internal data without it being exposed. And so huge amount of effort going into this. It will take some time. I don't know if it's going to work or not. Could you do the same with RNA? I asked Riju Dais, who knows about these things. And he said, to get enough training data, you probably want 10 to 100,000 structures costing somewhere between 100 million and 1 billion. So that's probably not going to be the answer there if there is an answer. You've got to find something else other than dumbly, I don't want to be rude to experimentalists, I was one once, generating more data. You might be able to do synthetic data for protein ligands. It's been suggested you might use molecular dynamics to get a training set, which I think is pretty cheeky for the molecular dynamics people, but we'll see. And there could be a possibility of bootstrapping. So a lot of things being tried. In some sense, a very simple idea, throwing

money at the problem sometimes might work. What else could you do? So the thing I like, which there's not much evidence for, but I'll tell you what there is, is combining the deep learning methods with classical methods. So there's decades of development of methods not competitive on their own with deep learning, but certainly not garbage. And so if you could combine the two, you might get a better performance. And we saw a hint of that in this CASP. This is for difficult complexes, the ones that are in those outliers for protein complexes, where you don't have evolutionary information defining the interface. This is a nano-body antigen complex, and you can see it's beautifully predicted. It's this one here, and the best method that did this structure is Koziakov-Vadger. You can see all of the other methods, ColabFold, two versions of alpha-3 and so on, do not work. So what did they do? They had an FFT physics-based method, which tries to search for poses. And so they ran that, they generated a set of poses, and then they used those as input to basically alpha-fold 3. And so you could regard it as a way of sampling, but it's a very intelligent way of sampling. It doesn't work every time yet. You can see there are still actually three failures. These are the set of targets we had for this class of problem. But I think there's a lot of hope here, and we saw a hint the other way around, do the physics afterwards for RNA. And so I know a lot of people are working on this, and I'd be surprised if there isn't something substantial in a year or two. Now, the third thing you could do, of course, is try a better algorithm. And there are a lot of ideas out there. So it could be the algorithm, it could be components, it could be the network setup. So lots, of course, you're familiar with this, convolution, graphs, transformers, and so on, represent atoms, residues, graphs, interaction types. All of these possibilities are being explored if you go to the literature. And so all you've got to do is read the literature, get a good idea, and apply it, and win the next CASP. The problem is that if you go to the literature for deep learning, you're just proteins now. In the last five years, there are just under 3,000 papers. So we could do a little quiz. I guess Gustavo's got a quiz of some sort later on, as to whether or not you could tell me what's in each one of these papers, even if you were a specialist. And of course, you can't. So we're overwhelmed with ideas, some of which are just ideas, some of which are really substantiated by what's in the papers. But notoriously, you cannot rely on the peer review of these papers to tell you what to believe. You need a different way of doing critical assessment. And here's where I flip to the other arrow. How can you use AI, in this first example, I'm going to get to a second one, how can you use AI to critically assess the papers? And so I don't have many slides in for this, but just to give you the idea, the idea is that we define a set of criteria we'd like to be fulfilled. And then we ask people, me, to try and fill in those criteria. And then we ask different AI tools, nothing fancy here. We use sort of three central AI tools for this work, very amateurish, to do the same thing and see how well you can do. So here's an example using actually just ChatDP with a lot of prompt engineering, of course, to try and look at it. It's an old paper. It's a diffusion docking, ligand docking paper. And the things we're looking at is what's the paper about? That's sort of easy. And there is a score for all this. Is this something which is interesting? Yes.

Are there new ideas here? Yes, there are nice ideas. Is the algorithm well described? Yes, but some of the stuff may be inaccessible. But it is on GitHub, so that's good. Is the data available? Well, no, not really. And there's some doubts. I've not shown it on here about the training-testing distinction. So there's only a one for that. Is there a server? No, maybe that doesn't matter. Is it innovative? Yes, it's very innovative. So the idea is you can run this against the whole of the literature, not do it by hand. And you come to a paper, you want to say, well, what does the AI, or AIs preferably, think about this paper, get a quick impression of how seriously the method has been tested, whether it's a good idea, a novel idea relative to the background. Now, that's, I think, would be useful, but it's not enough. What you'd want on top of that, I think, is some kind of dynamic ontology. What's out there, what's been looked at, how much, with what sort of methods. And so this is sort of Mickey Mouse demonstration of principle on the next slide. I took the CASP abstracts. There's about 150 of them. They're very rough. We asked people to submit abstracts. They're not in any way peer-reviewed or standard or anything. And said, what methods are in these abstracts? And so this is a simple breakdown of that. It's sort of boring sometimes. Look, a lot of people use deep learning. What a surprise. But on the other hand, a lot of people use different sorts of sampling. And that's interesting because you know sampling is helping in a lot of these problems. So that's your top level of the ontology. Now you say, okay, what do you mean sampling? I would have a table in here if I'd put the slide in. What sorts of sampling are people doing? And then you could cross-correlate that. Of course, the abstracts don't contain the results. But you could cross-correlate that with the results and say, this sort of sampling tends to give a better result than this other sort of sampling. So that's the concept. It's very much a concept, but I think it's extremely doable. Okay, so this is all within computational structural biology. But I think a lot of us, we've just heard the virtual cell talk, are interested in other sorts of assessment. And so one I've gotten very interested in is the problem of critical assessment in preclinical biology. So this is a bit of a jump. Let me introduce it with a slide. If there's a so-called reproducibility crisis in that field, and there's been careful studies to try and see how big a problem that is. So the idea is you take respectable papers, you look what experiments are in them, what the conclusions were, and then you try very hard, not just, you know, tried it didn't work stuff. Can I reproduce the reported result? So this is from this very nice large-scale study 2021, and you're looking at two things here. How well does the size of the effect which is reported reproduce? Let's say it's a five-fold change in expression of some gene. This is cancer stuff in some cancer cell. Is that reproduced? And secondly, is it significant by this traditional, very poor 0.05 probability criterion? So how much is that reproduced? So along the bottom here, you've got the reported cell size in log 2, so 2 to the 5 here. This is the replicated size. Unlike the RNA, the points should be on the diagonal here, and you can see they're not. They're more or less along, nothing happened. So startlingly, in these data, the average reported effect size in the reproduction is 20% of the original. And even if the statistical significance holds up, that takes you from biologically significant very often to biologically

irrelevant. And then the colors here, the blue, you do replicate with significance, the orange, not surprisingly given where they are, you've lost your 0.05 significance. A few things actually flip. Here's something with quite a big change, but the opposite sign to what it was originally, and confidently not the right way around. So that's the problem this field has. And I would argue that's having a big impact on how well we can use all of these data. There's lovely, rich data out there, but nobody knows what to trust. And I'll show you one example of that to make the point. We're interested in one particular variant that affects risk of Alzheimer's disease. Go into a lot of specifics, but it's a missent single base change in this gene ApoE. So this is called ApoE4. If you look at the risk of getting Alzheimer's, most people are E3E3. And so this is for different alleles, the chance of not having Alzheimer's as a function of age. The black line is E3E3. The far red line here is E4E4. So you can see just there's a huge increase in risk here. Now a couple of points. If you understood why that increase was there, you could perhaps design prophylactics just for people with this variant. You might also be able to design prophylactics for people as you watch biomarkers over the years and say, because I know what's going on here, if I give you this drug, I can hold off Alzheimer's until you're dead, sort of thing. So that's what you would like to do. It's not an original thought. There are 7,000 papers on the E4 variant and risk of Alzheimer's. So it's worse than deep learning. So, oh, by the way, these are my colleagues on this project. So we did this first by hand. We sort of went through the 7,000 papers and we said, we're thinking in mechanism terms here. What mechanisms have been proposed to explain this relationship? Now, I think what we know about other genetic variants and disease is that typically there's one mechanism. These systems are very robust, of course, and so usually just changing one base may give you one way to some kind of disease phenotype. But here, if you go to the literature, you find, it could be three. If you go to the literature and look, this is us now, here are 30, actually I think there's only 28 on the plot, but it goes on beyond 30, different biologically discrete mechanisms which are in the literature supported by data. So there's not speculation here. There's a paper, it says experiment, experiment, experiment, conclusion. And what are you going to believe here? At least 90%, probably more, are wrong in some serious sense. How do you decide what's wrong and what's right? By the way, what I'm going to tell you by hand works. So you see there are big grances here. This is sort of classical amyloid stuff. There's a lot of role for lipids, perhaps. There's this other protein involved in Alzheimer's called tau. So we've been able pretty much by hand to eliminate the tau lot. We're very doubtful about the lipid lot. We're absolutely sure there is some mechanism involving amyloid. Believe you me, that's not an obvious conclusion in the field. And that's from doing the sort of thing I'm about to describe. What we do is to go the same kind of line as with the AI for looking at papers. We have a set of structured evidence criteria, and hopefully, yes, here's an outline of that. So they fall into four classes. There's often extrapolation involved in these experiments. So how valid is the extrapolation? Are you in a different organism, in vitro, in a cell, whatever. Statistics are where most of these papers fall down. I think that, not disrespecting my

biology colleagues, they often don't have a good grasp of statistics, and the analysis is not valid, and that's where you get that sort of effect I showed you on the non-reproduction. On the other hand, some of these have been reproduced by different people. Some of them fit very well into other kinds of considerations of the biology. So you can plod through these rules by hand. You can score them. You can come up with a confidence. But you can also do it, again, with the AI tools. And I guess I'm about to run out of time. So this is a summary of trying to do that with this particular problem. There are different problems. Can you find the papers? Can you get the key information? Can you extract key results? Can you deduce the mechanism implications? Can you build a full mechanism? Can you critically assess the experimental results? Can you propose missing experiments? How easy is it to use? Generally low. In our hands, it takes forever, but this is obviously clearly cleanable, upable, and automatable on a large scale. I put probably better than people here. Actually, I'm convinced now it's a lot better. This is an old slide. So in my mind, we could take the whole of the biomedical literature, we could search for experiments, we could assess the reliability of the experiments, we could extract mechanism automatically, we could make a mechanism knowledge graph and find new mechanisms. That's the sort of vision here. It's obviously a ways to go with development, but I think that's really the future. And here comes the chair. I'm done. One minute, I finished. I don't know what to say for a minute. Seriously, I finished. So we are actually one of our brains. I see in the article.